

Options :

6406534468499. ✖ The page immediately shows "Fetched Data from API!" with no delay.

6406534468500. ✔ The message "Loading data..." is shown for ~1 second, then replaced with "Fetched Data from API!".

6406534468501. ✖ Both "Loading data..." and "Fetched Data from API!" appear together.

6406534468502. ✖ Nothing happens because `v-show` and `v-if` cannot work together.

Question Number : 50 Question Id : 6406531337325 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose the button is clicked multiple times quickly, the message area flickers unexpectedly. How can you improve the code to ensure proper async behavior and prevent overlapping fetches?

Options :

6406534468503. ✖ Add `v-else` after `v-if` instead of `v-show`.

6406534468504. ✖ Replace
`await this.simulateApiCall().then(...);` with
`const res = await this.simulateApiCall(); this.message = res;`

6406534468505. ✔ Add a check to disable the button while loading;
`<button @click="fetchData" :disabled="loading">Fetch</button>`

6406534468506. ✖ Use `v-html` instead of `v-show` for message rendering.

Game Theory and Strategy

Section Id :	64065394924
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory

Number of Questions :	6
Number of Questions to be attempted :	6
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653217603
Question Shuffling Allowed :	No

Question Number : 51 Question Id : 6406531337326 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534468507.  YES

6406534468508.  NO

Sub-Section Number :	2
Sub-Section Id :	640653217604
Question Shuffling Allowed :	No

Question Id : 6406531337327 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (52 to 54)
Question Label : Comprehension

(Bayesian Game) Consider a two-player game in which the payoffs, which depend on θ , and actions are as in the following table:

	$\theta = 0$			$\theta = 1$	
	L	R		L	R
U	1, -1	-1, 1	U	1, 1	-1, -1
D	-1, 1	1, -1	D	-1, 1	1, -1

Where $\Pr(\theta = 0) = \Pr(\theta = 1) = \frac{1}{2}$. Player 2 knows exactly what value θ has, whereas player 1 doesn't know.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 52 Question Id : 6406531337328 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Under Bayesian Nash Eq., player 1 will play

Options :

6406534468509. ✓ U

6406534468510. ✗ D

Question Number : 53 Question Id : 6406531337329 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Under Bayesian Nash Eq., player 2, who knows that $\theta = 0$ will play

Options :

6406534468511. ✗ L

6406534468512. ✓ R

Question Number : 54 Question Id : 6406531337330 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Under Bayesian Nash Eq., player 2, who knows that $\theta = 1$ will play

Options :

6406534468513. ✓ L

6406534468514. ✗ R

Sub-Section Number :

3

Sub-Section Id :

640653217605

Question Shuffling Allowed :

No

Question Id : 6406531337331 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (55 to 62)

Question Label : Comprehension

Each one of two political parties can choose to buy time on commercial radio shows to broadcast negative ad campaigns against their rival. These choices are made simultaneously. Due to government regulation, it is forbidden to buy more than 2 hours of negative campaign time, so that each party cannot choose an amount of negative campaigning above 2 hours. Given a pair of choices (a_1, a_2) , the payoff of party i is given by the following function: $v_i(a_1, a_2) = a_i - 2a_j + a_i a_j - (a_j)^2$. Write the normal form representation of this game and answer the given sub-questions.

Sub questions

Question Number : 55 Question Id : 6406531337332 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If player 1 chooses 2 hours as his action, the best response of player 2 will be

Options :

6406534468515. ✓ 1.5

6406534468516. ✗ 1

6406534468517. ✗ 1.25

6406534468518. ✗ 0

Question Number : 56 Question Id : 6406531337333 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How much negative campaign time will each player choose under a symmetric and unique Nash equilibrium?

Options :

6406534468519. ✗ 1.5

6406534468520. ✓ 1

6406534468521. ✗ 1.25

6406534468522. ✗ 0

Question Number : 57 Question Id : 6406531337334 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Equilibrium payoff of each player will be

Options :

6406534468523. ✗ -1.5

6406534468524. ✗ 1

6406534468525. ✓ -1

6406534468526. ✗ 0

Question Number : 58 Question Id : 6406531337335 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

If the parties could sign a binding agreement on how much to campaign, what levels would they choose?

$a_1 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 59 Question Id : 6406531337336 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

If the parties could sign a binding agreement on how much to campaign, what levels would they choose?

$a_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 60 Question Id : 6406531337337 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely and the parties cooperate and sign a binding agreement on how much to campaign.

What will be the payoff of each player in each stage game if they stick to this agreement

Options :

6406534468529. ✖ -1.5

6406534468530. ✖ 1

6406534468531. ✖ -1

6406534468532. ✓ 0

Question Number : 61 Question Id : 6406531337338 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely and the parties cooperate and sign a binding agreement on how much to campaign.

What should be the level of campaign
a player should choose if wants to deviate
from the agreement to get the payoff of $\frac{1}{4}$

Options :

6406534468533. ✖ 1

6406534468534. ✓ $\frac{1}{2}$

6406534468535. ✖ $\frac{1}{3}$

6406534468536. ✖ $\frac{1}{4}$

Question Number : 62 Question Id : 6406531337339 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now assume that this game is repeated infinitely and the parties cooperate and sign a binding agreement on how much to campaign.

For which discount factors $\delta \in (0, 1)$ can this
agreement cannot be supported as a subgame
perfect equilibrium of the infinitely repeated game?

Options :

6406534468537. ✖ $\delta \geq \frac{1}{5}$

6406534468538. ✓ $\delta \leq \frac{1}{5}$

6406534468539. ✖ $\delta \geq \frac{1}{3}$

$$\delta \leq \frac{1}{3}$$

6406534468540. ✖

Sub-Section Number : 4
 Sub-Section Id : 640653217606
 Question Shuffling Allowed : No

Question Id : 6406531337340 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
 Question Numbers : (63 to 64)
 Question Label : Comprehension
 Consider the following game matrix:

		Player 2	
		Slow	Fast
Player 1	Slow	6,6	0,2
	Fast	2,0	1,1

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 63 Question Id : 6406531337341 Question Type : MCQ
 Correct Marks : 1

Question Label : Multiple Choice Question
 Choose the correct alternative(s)

Options :

6406534468541. ✔ (Slow, Slow) is an NE
 6406534468542. ✖ (Slow, Fast) is an NE
 6406534468543. ✖ Both (Slow, Slow) is an NE & (Slow, Fast) is an NE
 6406534468544. ✖ None

Question Number : 64 Question Id : 6406531337342 Question Type : MCQ
 Correct Marks : 1

Question Label : Multiple Choice Question
 ESS in this game is (are)

Options :

6406534468545. ✖ Slow
 6406534468546. ✖ Fast
 6406534468547. ✔ Both Slow & Fast

6406534468548. ✖ None

Sub-Section Number : 5
Sub-Section Id : 640653217607
Question Shuffling Allowed : No

Question Id : 6406531337343 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (65 to 70)

Question Label : Comprehension

Consider the example of cascade discussed in the lecture but with a modification that now the urn is containing either 4 blue balls and 2 red ball or 4 red balls and 2 blue ball with equal probability. That simply means that the urn is equally likely to be blue-majority urn or red-majority urn. We assume everyone hears what the previous individuals report and get their own private signal by randomly drawing one ball from the urn.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 65 Question Id : 6406531337344 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Suppose one ball is picked from the red majority urn. What is the probability that the ball is blue?

Options :

6406534468549. ✔ 1/3

6406534468550. ✖ 3/4

6406534468551. ✖ 1/4

6406534468552. ✖ 2/3

Question Number : 66 Question Id : 6406531337345 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is the probability that the urn is majority-red if the first student draws a blue ball?

Options :

6406534468553. ✖ 1/5

6406534468554. ✖ 4/5

6406534468555. ✖ 2/3

6406534468556. ✔ 1/3

Question Number : 67 Question Id : 6406531337346 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the probability of getting this sequence if the urn is a blue-majority urn?

Options :

6406534468557. ✖ 5/27

6406534468558. ✔ 4/27

6406534468559. ✖ 16/125

6406534468560. ✖ 9/125

Question Number : 68 Question Id : 6406531337347 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the probability of getting this sequence if the urn is a red-majority urn?

Options :

6406534468561. ✔ 2/27

6406534468562. ✖ 5/27

6406534468563. ✖ 5/216

6406534468564. ✖ 7/216

Question Number : 69 Question Id : 6406531337348 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the total probability of getting this sequence?

Options :

6406534468565. ✖ 1/25

6406534468566. ✖ 2/25

6406534468567. ✔ 1/9

6406534468568. ✖ 2/9

Question Number : 70 Question Id : 6406531337349 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Blue, Blue, Red).

What is the probability of urn being red-majority in this scenario?

Options :

6406534468569. ✖ 2/3

6406534468570. ✖ 1/25

6406534468571. ✖ 4/25

6406534468572. ✔ 1/3

Sub-Section Number : 6
Sub-Section Id : 640653217608
Question Shuffling Allowed : No

Question Id : 6406531337350 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (71 to 75)
Question Label : Comprehension
Given the following preference structure, find a stable matching system using the Gale–Shapley algorithm when the men propose

Women: A, B, C, D, E.						Men: a, b, c.					
	A	B	C	D	E		A	B	C	D	E
a	1	2	1	2	1	a	5	1	2	3	4
b	3	3	2	1	2	b	4	1	2	5	3
c	2	1	3	3	3	c	5	4	1	3	2

Based on the above data, answer the given subquestions.
Sub questions

Question Number : 71 Question Id : 6406531337351 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question
Ms. A will be paired with

- Options :
- 6406534468573. ✖ a
 - 6406534468574. ✖ b
 - 6406534468575. ✖ c
 - 6406534468576. ✔ None

Question Number : 72 Question Id : 6406531337352 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question
Ms. B will be paired with

- Options :
- 6406534468577. ✔ a
 - 6406534468578. ✖ b
 - 6406534468579. ✖ c